Department of Computer Science

Banasthali Vidyapith

Software Requirement Specifications (SRS) and Software Design Specifications (SDS)

Inventory Management System

CSD026

Guided by:

Mainaz Faridi Mam

Submitted By:

Palak Patodi	2316496
Somya Jain	2316571
Soumya Arora	2316572
Suhani Jain	2316576
Vanshree	2316602

Software Requirements Specification

for

INVENTORY MANAGEMENT SYSTEM

Version 1.0 approved

Table of Contents

1. Introduction

1.1 Purpose

The purpose of this Software Requirements Specification (SRS) document is to define the functional and non-functional requirements for the Inventory Management Application (IMA) developed for the Faculty of Mathematics and Computing. This desktop application will manage and track departmental inventory across the following areas: Stationery, Furniture ,Computer set and miscellaneous items like Mop, Broom etc . The application will streamline record-keeping, allocation of items, and monitoring of stock availability. Authorized users include 2–3 office staff members responsible for inventory operations, and the HOD who has administrative access. This document pertains specifically to the core software system and its scope as defined within this project release.

1.2 Document Conventions

This SRS follows standard IEEE guidelines for documenting software requirements. Requirements are presented in a structured format with numbered sections. Tables and entities are described at a high level. Technical details regarding implementation are limited to the technologies chosen (Java, JavaFX, and SQL). All requirement statements are given equal priority unless otherwise specified.

1.3 Intended Audience and Reading Suggestions

The intended audience of this SRS includes:

Developers: to understand functional and backend requirements when implementing the system.

Project Managers: to verify scope, workflows, and deliverables.

Inventory Managers: to understand how the system supports departmental processes.

HOD and Dean(Administrator): to review administrative control and reporting features.

Testers: to derive test cases and validate that requirements are fulfilled.

Readers are advised to first review the Introduction and Product Scope for context, followed by functional requirements. Developers and testers should focus on the system features and database structures, while management may focus on use cases and constraints.

1.4 Product Scope

The IMA will provide a centralized digital platform for stationery, furniture, computer sets and miscellaneous inventory management for faculty of Mathematics and Computing. The core objectives are to:

- Track real-time availability of products, avoiding stockouts or overstocking.
- Ensure materials/products are available when needed to meet customer or production requirements.
- Streamline purchasing, storage, and distribution processes.
- Maintain clear records of stock movement, ownership, and usage.
- Automated calculations minimize human errors in stock counts and data entry.
- Users can access analytics, and reports instantly.
- Reduced paper, storage, and administrative costs; less wastage due to better control.
- Digital records are searchable; no need to shuffle through piles of registers.
- IMA can connect with accounting, sales, and procurement systems for smooth workflow.
- As inventory grows, digital systems handle large volumes easily without proportional manpower increase.
- Easier reporting and accurate logs simplify internal and external audits.

The proposed system will be built using Java and JavaFX for the front-end interface and Java with SQL for backend and database operations.

1.5 References

IEEE Recommended Practice for Software Requirements Specifications, IEEE Std 830-1998.
JavaFX Documentation, Oracle, latest release.
Java SE Platform Specification, Oracle, version 17 (LTS).
MySQL Official Reference Manual, Oracle, latest edition.
Departmental guidelines for inventory tracking, faculty of Mathematics and Computing (internal document).

2. Overall Description

2.1 Product Perspective

The Inventory Management Application (IMA) is a new, standalone software application developed for the faculty of Mathematics and Computing to replace manual, paper-based inventory tracking. It is not part of a larger product family but functions independently, without requiring integration with external enterprise tools.

The application automates purchase tracking, allocation management, availability updates, and reporting. It directly interacts with an SQL database for data storage and retrieval.

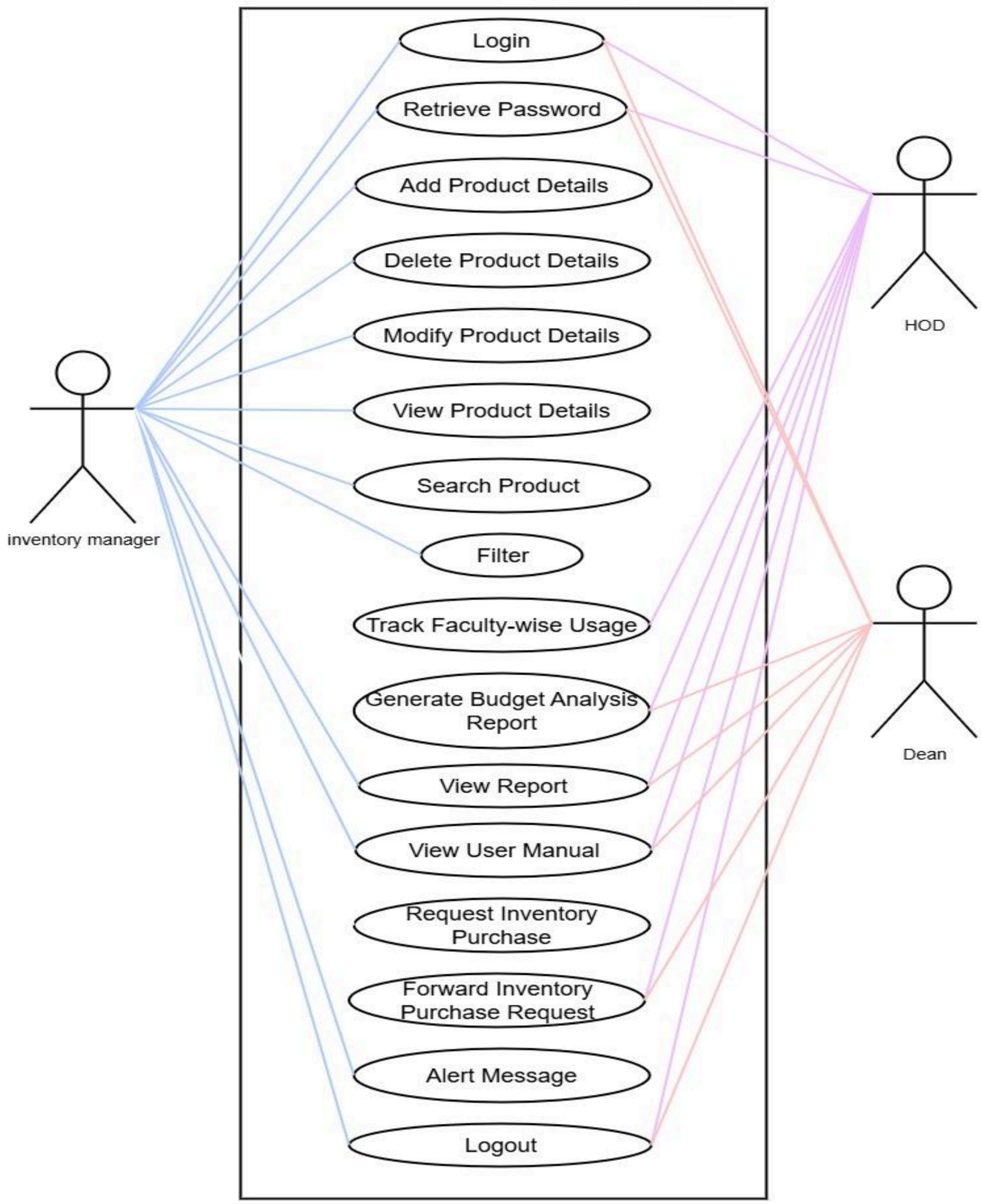
2.2 Product Functions

The standalone application, now with a unified database structure, will perform the following high-level functions:

- Authentication & Access Control: Secure logins for Inventory Managers,HOD and the Dean.
- Product Management: A ***Product Table*** will maintain all items, and a ***Product Type Table*** will differentiate between stationery, furniture, computer set and other product categories. The system will track vendor/purchase records, staff and room allocations, and usage. This includes historical records, current availability, and usage tracking.
- Procurement & Suppliers: A ***Supplier Table*** will manage seller and vendor records, and a ***Bill Table*** will track purchase bills and quantities for all procured items.
- Issuance & Returns**:** An ***Issue Table*** will record all items issued to staff and departments, providing detailed tracking of usage. The system will also include a ***Return Table*** to manage and track all items returned to inventory or to supplier, ensuring accurate stock levels and a complete usage history.
- Item Request Management: Inventory Managers will have the functionality to raise requests for new items or to replenish existing stock through mail.
- Reporting and Monitoring: The system will provide comprehensive reporting and monitoring capabilities through a user-friendly interface, offering summaries for all users, usage pattern analysis, real-time availability checks, and detailed budget analysis. To enhance clarity and understanding, all reports will be supported by dynamic graphs and visualizations.
- Automation: The application will automate and replace manual, paper-based record-keeping, reducing errors and improving data accessibility.
- Oversight: The HOD and Dean will have high-level oversight, reporting capabilities, and administrative rights.Requests for items made by managers must be approved by the HOD before an order can be placed.If the items are computer sets or a large order, the Dean also needs to approve the request.This project aligns with the

department's strategy to improve transparency, accountability, and efficiency in handling resources. By consolidating data into a common database, the system will provide a more streamlined and integrated approach to managing all inventory.

Use Case Diagram and Templates:



Uname	Login
Uno.	01
Actor	Inventory Manager,Hod,Dean
Pre-condition	User must have a registered account.
Post-condition	User is authenticated and logged in.System is ready for subsequent use cases.
Normal flow	User enters valid credentials, system authenticates and then user is redirected to the main interface where available tasks are displayed.
Abnormal flow	Invalid credentials shows an error and allow retries. User can select "forgot password" to reset credentials.

Uname	Retrieve Password
Uno.	02
Actor	Inventory Manager, HOD, Dean
Pre-condition	User is on the login screen. User must have a registered account.
Post-condition	A password reset link or code is sent to the user. User can create a new password
Normal flow	User clicks "forgot password", provides identification and receives a reset link or code to change their password
Abnormal flow	Invalid user details show an error message is displayed.

Uname	Add Product Details
Uno.	03
Actor	Inventory Manager
Pre-condition	User is logged in as Inventory Manager. "Add Product" form is accessible.
Post-condition	New product is saved in the system and listed in the inventory database.
Normal flow	User opens the "Add Product" form, enters required product details, and submits the form; system saves the new product successfully.
Abnormal flow	Missing or invalid input shows an error; duplicate product code shows a warning; system failure during save displays an error message.

Uname	Delete Product Details
Uno.	04
Actor	Inventory Manager
Pre-condition	User is logged in as Inventory Manager, Product exists in the inventory.
Post-condition	Selected product is removed from the inventory records.
Normal flow	User selects a product from the inventory list, confirms deletion, and the system removes the product from the database.
Abnormal flow	If product doesn't exist, an error is shown; deletion is canceled if user declines confirmation; system error during deletion shows a failure message.

Uname	Modify Product Details
Uno.	05
Actor	Inventory Manager
Pre-condition	User is logged in as Inventory Manager. Product exists in the inventory.
Post-condition	Updated product details are saved and reflected in the inventory records.
Normal flow	User selects a product, edits the necessary fields, and submits the changes; system validates and saves the updated product details.
Abnormal flow	Invalid inputs show an error; if the product is not found, a message is displayed; save failure due to system error shows an error message.

Uname	View Product Details
Uno.	06
Actor	Inventory Manager
Pre-condition	User is logged in, Product records exist in the system.
Post-condition	Product details are displayed to the user.
Normal flow	User navigates to the product list, selects a product, and the system displays its details.
Abnormal flow	If no products are found, the system shows a "No records available" message; system error while fetching data displays an error message

Uname	Search Product
Uno.	07
Actor	Inventory Manager
Pre-condition	User is logged in, Product data exists in the system.
Post-condition	Matching product(s) are displayed based on the entered search criteria.
Normal flow	User enters a keyword (e.g., product name or code), system searches the inventory, and displays the matching results.
Abnormal flow	If no matching products are found, the system displays “No results found”; if search fails due to system error, an error message is shown.

Uname	Filter Products
Uno.	08
Actor	Inventory Manager
Pre-condition	User is logged in, Product data is available.
Post-condition	Filtered list of products is displayed based on selected criteria.
Normal flow	User selects filter criteria (e.g., category, availability), system processes the input, and displays the filtered product list.
Abnormal flow	If no products match the filter, a "No results found" message is shown; if the system fails to apply filters, an error is displayed.

Uname	Track Faculty-Wise Usage
Uno.	09
Actor	HOD
Pre-condition	User is logged in, Usage data by faculty exists in the system.
Post-condition	A detailed or summary view/report of usage per faculty is displayed or exported.
Normal flow	User selects the "Track Faculty-Wise Usage" option, applies filters (e.g., date range or faculty name), and the system displays the relevant usage records.
Abnormal flow	If no usage data is found for the selected faculty, a "No data available" message is shown; system errors display a failure message.

Uname	Generate Budget Analysis Report
Uno.	10
Actor	HOD, Dean
Pre-condition	User is logged in, Budget and transaction data exists in the system.
Post-condition	Budget report is generated and displayed or exported in the selected format.
Normal flow	User selects "Generate Budget Analysis Report", applies filters (e.g., date range, department), and system generates the report for viewing or export.
Abnormal flow	If no data is available for the selected criteria, a "No data found" message is shown; system errors during generation display an error message.

Uname	View Report
Uno.	11
Actor	Inventory Manager, HOD, Dean
Pre-condition	User is logged in. Report data exists in the system.
Post-condition	Selected report is displayed on screen or made available for download/export.
Normal flow	User navigates to the reports section, selects a report type, and the system displays the requested report.
Abnormal flow	User navigates to the reports section, selects a report type, and the system displays the requested report.

Uname	View User Manual
Uno.	12
Actor	Inventory Manager, HOD, Dean
Pre-condition	User is logged in.User manual is available in the system.
Post-condition	User manual is displayed for viewing or downloaded.
Normal flow	User clicks on "User Manual" or "Help", and the system opens the manual in a viewer or downloads the document.
Abnormal flow	If the manual file is missing or corrupted, an error message is shown; system error prevents access and displays a failure message.

Uname	Request Inventory Purchase
Uno.	13
Actor	Inventory Manager
Pre-condition	User is logged in as Inventory Manager.Access to the purchase request form is available.
Post-condition	Purchase request is submitted and saved for review by HOD and Dean.
Normal flow	User fills out the purchase request form with item details and submits it; system saves the request and notifies the HOD for approval.
Abnormal flow	If required fields are missing or invalid, an error is shown; if system fails to submit the request, an error message is displayed.

Uname	Forward Inventory Purchase Request
Uno.	14
Actor	HOD, Dean
Pre-condition	User is logged in as HOD or Dean.A pending purchase request is available for review.
Post-condition	Request is forwarded to the next level (HOD → Dean or Dean → Final Approval).
Normal flow	HOD reviews the purchase request and forwards it to the Dean; Dean reviews and forwards it for final processing or approval.
Abnormal flow	If request details are incomplete, forwarding is blocked and an error is shown; if system error occurs during forwarding, a failure message is displayed.

Uname	Alert Message
Uno.	15
Actor	Inventory Manager
Pre-condition	The user is logged in, and an alert is triggered because the product's minimum quantity threshold has been reached.
Post-condition	Alert message is shown to the user and User is aware and may act on the message.
Normal flow	System detects a predefined condition (e.g., low inventory), generates an alert, and displays it on the user's screen or sends it as a notification.
Abnormal flow	If alert fails to display due to a system error, a fallback notification method i.e. retry (The system attempts again to display the alert after a short delay.) is used.

Uname	Logout
Uno.	16
Actor	Inventory Manager, HOD, Dean
Pre-condition	User is logged in
Post-condition	User session is terminated and User is redirected to the login screen.
Normal flow	User selects the logout option; system terminates the session and redirects to the login screen.
Abnormal flow	If logout fails due to system error, an error message is displayed, and the user remains logged in.

2.3 User Classes and Characteristics

Inventory Managers: End users who manage procurement and allocation through the application, while Inventory Managers also receive automatically generated monthly inventory reports on their dashboard to support both daily management and periodic operational reviews.
HOD and Dean(Administrator): Occasional user overseeing reports and handling administrative monitoring along with request approvals.

2.4 Operating Environment

Hardware: Standard departmental PCs (at least 4 GB RAM or higher as per data, Mid-range CPU).
Operating System: Windows 10 or higher (Linux optional with Java setup).
Frontend: JavaFX-based GUI.
Backend: Java-based logic with SQL database.

Runtime: Java SE version 17 LTS or higher.

2.5 Design and Implementation Constraints

- The GUI only supports the English language.
- Authentication restricted to authorized office staff members, HOD and the Dean.
- Scalability limited to departmental-level data volume (for faculty of Mathematics and Computing).

2.6 User Documentation

The delivered package will include a User Manual (PDF) and a Guide describing advanced configurations and reporting.

2.7 Assumptions and Dependencies

Users will have access to computers with Java Runtime and the database setup already in place.
Data will be entered accurately by office staff members.
The application depends on:

- Java SE environment
- SQL database running locally or on the department server
- Secure credential handling by the department.

If the SQL database service is unavailable, the application's functions will be impacted.

3. External Interface Requirements

3.1 User Interfaces

The application provides a graphical user interface (GUI) implemented using JavaFX, designed to be intuitive and accessible for office staff. The GUI follows a consistent style guide with a clean, minimalistic layout and uses:

- Standard window controls (minimize, maximize, close).
- Navigation menus for major modules: Product Management, Reporting, and Item Request Management.
- Forms with labeled fields for inputting vendor details, purchase data, staff allocations, and issue logs.

- A dedicated form for managers to create and submit new item requests, specifying the product type, quantity, and reason. Submitting a request will automatically generate and send an email to the Dean for approval.
- Table views for displaying lists and records with searching and filtering capabilities. This includes a dedicated ***Return Table*** for managers to record and track returned items.
- Common action buttons across screens: Add, Edit, Delete, Save, Search, and Refresh.
- Validation error messages clearly displayed near form fields, with standardized alert dialogs for critical errors.
- Keyboard shortcuts for common operations (e.g., Ctrl+S for Save, Ctrl+F for Search).
- Reports and Dashboards: The application will include a dedicated reports section with dynamic dashboards tailored to each user. Inventory managers will see daily, monthly, and yearly reports summarizing key inventory metrics, while HOD and the Dean will have access to a Budget Analysis Report and a Faculty Usage Report to track spending and resource allocation at a strategic level. These reports will be a mix of automatically generated summaries and filterable views, providing clear insights into operations, finances, and usage patterns.
- Screens are designed to accommodate standard display resolutions starting from 1024x768 pixels. Accessibility considerations include legible font sizes and color contrast compliant with common standards.

3.2 Hardware Interfaces

As a standalone desktop application, the application interfaces directly with the hardware through:

- Standard input devices: processor, CPU, keyboard and mouse for user interaction.
- Display device: computer monitor supporting at least 1024x768 screen resolution.
- Storage devices: local or networked storage hosting the SQL database server. Hard disk and RAM (4 GB or more).

No specialized hardware devices or sensors are required. The application communicates with the database server over the local network or local host machine.

3.3 Software Interfaces

The application depends on the following software interfaces:

- Java Runtime Environment (JRE), version 17 LTS or higher: The application requires a compliant Java SE runtime environment for execution.
- JavaFX Libraries: For constructing the graphical user interface components.

- SQL Database Server: The backend database can be MySQL (version 8.0 or higher). The application uses JDBC (Java Database Connectivity) API to interact with the database.
- Operating System: Primarily Windows 10 or above; Linux-compatible distributions that support Java and required databases are also supported.

The application does not require integration with external third-party software or web services. All data sharing occurs internally between the Java application and the SQL database.

3.4 Communications Interfaces

The standalone application does not require internet or external network communication protocols. Communication is limited to:

- Local network communication between the application and the database server using TCP/IP over standard database ports (e.g., 3306 for MySQL).
- No communication encryption is implemented beyond the database connection security protocols (e.g., SSL/TLS if configured on the SQL server).
- Data transfer rates depend on the local network speed but are expected to be minimal given the application's lightweight query nature.

Synchronization mechanisms rely on the database transaction management to maintain data integrity in multi-user scenarios.

4. System Features

4.1 Login

4.1.1 Description and Priority

This feature manages secure login for Inventory Managers, HOD and the Dean, ensuring role-based access. It is of High priority as it protects sensitive inventory data and enforces user privileges.

4.1.2 Stimulus/Response Sequences

The user launches the application and is prompted to enter a username and password. The application validates credentials against stored user data.

- If credentials are valid, the user is granted access to modules based on their role (Manager or Administrator).
- If credentials are invalid, an error message is displayed, and the user is prompted to re-enter their credentials.
- The login screen shall also include a "Forgot Password" link. Upon clicking, the user will be guided through a process to reset their password.

4.1.3 Functional Requirements

- REQ-1: The application shall require users to input a valid username and password at login.
- REQ-2: The application shall restrict access to inventory management functionalities until successful authentication.
- REQ-3: The application shall support three user roles: Inventory Manager, HOD and the Dean.
- REQ-4: The application shall display a distinct interface layout and enable/disable features based on the user's role.
- REQ-5: The application shall lock out users after three unsuccessful login attempts and display a warning message.
- REQ-6: The application shall provide an option for authenticated users to change their password.

4.2 Product Management

4.2.1 Description and Priority

This core feature manages all product records, purchases, and allocations. The system will use a common database structure, with a ProductType Table to differentiate between stationery, furniture, computer sets, and miscellaneous items. This is a High priority function for tracking all inventory from procurement to assignment.

4.2.2 Stimulus/Response Sequences

- A Manager selects the Product Management module.
- The application displays forms for details regarding supplier records, purchase bills, and item issuance.
- The manager inputs new purchase details (supplier, bill number, date, product type, quantity) for items that have already been procured.
- The manager also has the option to raise a request for new items or to replenish existing stock that needs to be ordered.
- The application will then route this request to the administrator (HOD/Dean) for approval.
- The manager inputs allocation details for an item (staff name, department, room, quantity).
- The application validates data and saves it to the database, updating the ***Product Table*** and ***Issue Table*** accordingly.
- The system provides a success confirmation or highlights errors.
- The application will also have a dedicated interface for managers to record returned items, updating the database to reflect the changes in stock.

4.2.3 Functional Requirements

- REQ-7: The application shall allow users to add, read, update, and delete Supplier and Bill records.

- REQ-8: The application shall allow users to add, read, update, and delete item issuance and return records for staff and departments.
- REQ-9: The application shall validate all input fields for required formats (e.g., date, numeric quantity).
- REQ-10: The application shall prevent an issued quantity from exceeding the available stock in the ***Product Table***.
- REQ-11: The application shall provide search and filter options on all ***Product, Supplier, Issue***, and ***Return tables***.
- REQ-12: Invalid inputs must trigger informative error messages guiding users to correct errors.
- REQ-13: The application shall provide a dedicated function for recording item returns, which automatically updates the available stock.
- REQ-14: The application shall provide a mechanism for managers to raise requests for new items or stock replenishment, which must be approved by an administrator before an order can be placed.

4.3 Reporting and Administrative Overview

4.3.1 Description and Priority

This feature provides robust reporting and monitoring capabilities, with visibility tailored to each user's role. It is a High priority for supporting strategic decision-making and operational transparency.

4.3.2 Stimulus/Response Sequences

- A user logs in and navigates to the "Reports" section. The application displays a dashboard with available reports based on their user role.
- For all Inventory Managers: The system automatically generates a set of reports on a daily, monthly, and yearly basis. These reports summarize key inventory metrics, including stock levels, allocations, and issues. Managers can view, search, and filter these reports for operational insights.
- For the HOD: The Dean can access and generate the Budget Analysis Report, which tracks spending against the budget. The Dean can also access the Faculty Usage Report to monitor resource allocation to faculty members.
- For Dean: Dean can also access the Reports.
- They can export reports in PDF format.
- The application validates all generated reports against database records to ensure accuracy.

4.3.3 Functional Requirements

- REQ-15: The application shall generate real-time reports showing inventory stock levels and allocation details for the Inventory managers, Dean and the HOD.
- REQ-16: The application shall allow exporting reports in PDF format.

- REQ-17: The application shall provide search and filter options within reports to analyze specific timeframes or items.
- REQ-18: The comprehensive reporting interface shall be accessible only to users with Administrative roles.
- REQ-19: The application shall log report generation activities for audit purposes.
- REQ-20: The application shall automatically generate and display a daily, monthly, yearly report for each Inventory Manager in their interface.
- REQ-21: The monthly report for Managers shall include summary information relevant to their operations and be available at the start of each month.

5. Other Nonfunctional Requirements

5.1 Performance Requirements

The application shall load the main dashboard within 3 seconds on standard departmental hardware.
Database queries for inventory retrieval and updates shall complete within 2 seconds for up to 10,000 records.
Concurrent access by up to 3 Inventory Managers, 1 HOD and 1 Dean shall not degrade performance noticeably.

5.2 Safety Requirements

User credentials shall be stored securely using salted hashing to prevent unauthorized access.
Role-based access control must ensure feature-level permissions are enforced consistently.
The application shall lock user accounts after 3 consecutive failed login attempts.
Sensitive data transmissions to the database shall be encrypted if the database supports secure connections.

5.3 Security Requirements

The GUI shall provide clear navigation and feedback messages to minimize user errors.
The application shall include tooltips and inline help for all input fields.
Keyboard shortcuts for common actions shall be provided to enhance efficiency.
User documentation, including a quick start guide and user manual, shall be accessible from the application.

5.4 Software Quality Attributes

The application shall provide data backup and recovery instructions to ensure minimal data loss.

All data modification operations shall support atomic transactions to maintain consistency.
The system shall handle application crashes gracefully and auto-save user progress when possible.

5.5 Business Rules

Code and documentation shall follow agreed departmental coding standards and be well-commented.
The database schema shall be normalized to ease future extensions or modifications.
Logs of critical operations (logins, data updates, report generation) shall be maintained for audit and troubleshooting.

5.6 Portability

The application shall run on Windows 10 or higher and Linux distributions that support the required Java runtime and database.
Deployment instructions shall be provided for setting up the application on new machines within the department.

6. Other Requirements

6.1 Database Requirements

The application shall use MySQL as the database management system to store all inventory, user, and transaction data.
Database tables will be normalized to reduce redundancy and maintain data integrity.
Backup and recovery procedures shall be established within MySQL to prevent data loss.
Indexing strategies will be implemented on key fields to ensure efficient query processing.
Secure JDBC connections and role-based access control shall be enforced to maintain data security.

6.2 Legal and Compliance Requirements

The system shall comply with relevant institutional data privacy and security policies.
Only authorized personnel are allowed access to the inventory management data and logs.
Comprehensive logs and audit trails will be maintained to ensure traceability and accountability.

6.3 Reuse Objectives

The project will leverage existing Java libraries and frameworks such as JavaFX and JDBC to promote maintainability.

The modular structure of the application will facilitate reuse and extension of components in future versions or related projects.

Appendix A: Glossary

Terms and their Definition:

Term	**Definition**
Application (IMA)	The Inventory Management Application software.
Inventory Manager	Office staff responsible for managing inventory data and allocations.
Dean and HOD	Administrative users with oversight privileges of the inventory.
Stationery	Office supplies and consumables tracked in the inventory system.
Furniture	Departmental furniture tracked in the inventory system.
Computer Sets	Computer hardware and peripherals tracked in the inventory system.
Miscellaneous Items	Other various items not categorized as stationery, furniture, or computer sets.
Product Type Table	A table used to categorize all items, differentiating between stationery, furniture, computer sets, and miscellaneous items.
Product Table	A common database table that maintains all inventory items, including historical and current data.
Supplier Table	A table that records information about sellers and vendors.
Bill / Invoice Table	A table that tracks purchase bills and quantities for all procured items.
Issue Table	A table recording details of all items issued to staff or departments.
Return Table	A table that records all items returned to vendors.
JDBC	Java Database Connectivity, API used for Java

	applications to communicate with SQL databases.
Role-Based Access	Security model granting permissions based on user roles (Inventory Manager or Dean).
SQL	Structured Query Language used for managing data in relational databases
Automated Reports	A set of standardized inventory summary reports generated automatically on a daily, monthly, and yearly basis.
Budget Analysis Report	A strategic report providing a financial overview of spending against the budget and comparisons between the last month and current month inventory lists.
Faculty Usage Report	A specialized report detailing item issuance to specific faculty members.

Chapter 2 SDS

TABLE OF CONTENTS

1. Introduction

This Software Design Specification (SDS) outlines the high-level and detailed design for the Inventory Management Application (IMA) for the faculty of Mathematics and Computing at Banasthali Vidyapith. The document is intended for the developers who will implement the system, project managers verifying the design, and testers creating test cases. It serves as a blueprint for the system's architecture, components, and data structures, translating the requirements from the Software Requirements Specification (SRS) into a technical design.

1.1 Purpose

The purpose of this SDS is to provide a comprehensive technical design for the IMA. It details the system's architecture, data design, user interface, and testing strategies to guide developers in the implementation process. The intended audience includes the software development team, project managers, and quality assurance personnel.

1.2 Scope

The software product to be produced is the IMA, a standalone desktop application for the faculty of Mathematics and Computing.The system's primary functions are to digitize and manage the department's stationery and furniture inventory. It will track procurement, allocation, and stock availability for both stationery and furniture items.The system will not integrate with external enterprise tools or web services. Its application is limited to departmental-level data volume. The key objectives are to streamline record-keeping, improve transparency, and support data-driven decision-making. The system is built using Java, JavaFX, and SQL

1.3 Definitions, Acronyms, and Abbreviations.

Terms and their Definition:

Term	Definition
Application (IMA)	The Inventory Management Application software.
Inventory Manager	Office staff responsible for managing inventory data and allocations.
Dean and HOD	Administrative users with oversight privileges of the inventory.
Stationery	Office supplies and consumables tracked in the inventory system.
Furniture	Departmental furniture tracked in the inventory system.
Computer Sets	Computer hardware and peripherals tracked in the inventory system.
Miscellaneous Items	Other various items not categorized as stationery, furniture, or computer sets.
Product Type Table	A table used to categorize all items, differentiating between stationery, furniture, computer sets, and miscellaneous items.
Product Table	A common database table that maintains all inventory items, including historical and current data.
Supplier Table	A table that records information about sellers and vendors.
Bill Table	A table that tracks purchase bills and quantities for all procured items.
Issue Table	A table recording details of all items issued to staff or departments.
Return Table	A table that records all items returned to vendors.
JDBC	Java Database Connectivity, API used for Java applications to communicate with SQL databases.
Role-Based Access	Security model granting permissions based on user roles (Inventory Manager or Dean).
SQL	Structured Query Language used for managing data in relational databases
Automated Reports	A set of standardized inventory summary reports generated automatically on a daily, monthly, and yearly basis.
Budget Analysis Report	A strategic report providing a financial overview of spending

	against the budget and comparisons between the last month and current month inventory lists.
Faculty Usage Report	A specialized report detailing item issuance to specific faculty members.

1.4 Overview

The remainder of this SDS is organized into several key sections. Section 2, System Architectural Design, provides a high-level overview of the system's components and their interactions. Section 3, Structure and Relationships, details the data design and relationships between system entities, including an Entity-Relationship (ER) model. Section 4, User Interface Design, describes the GUI layout and objects. Section 5, Types of Tests, outlines the different testing phases. Finally, Section 6 provides a list of references.

2. System Architectural Design

2.1 High-level Design Overview

IMA is a new, standalone software application that replaces manual inventory tracking in which all the modules interact directly with an SQL Database for all data storage and retrieval.

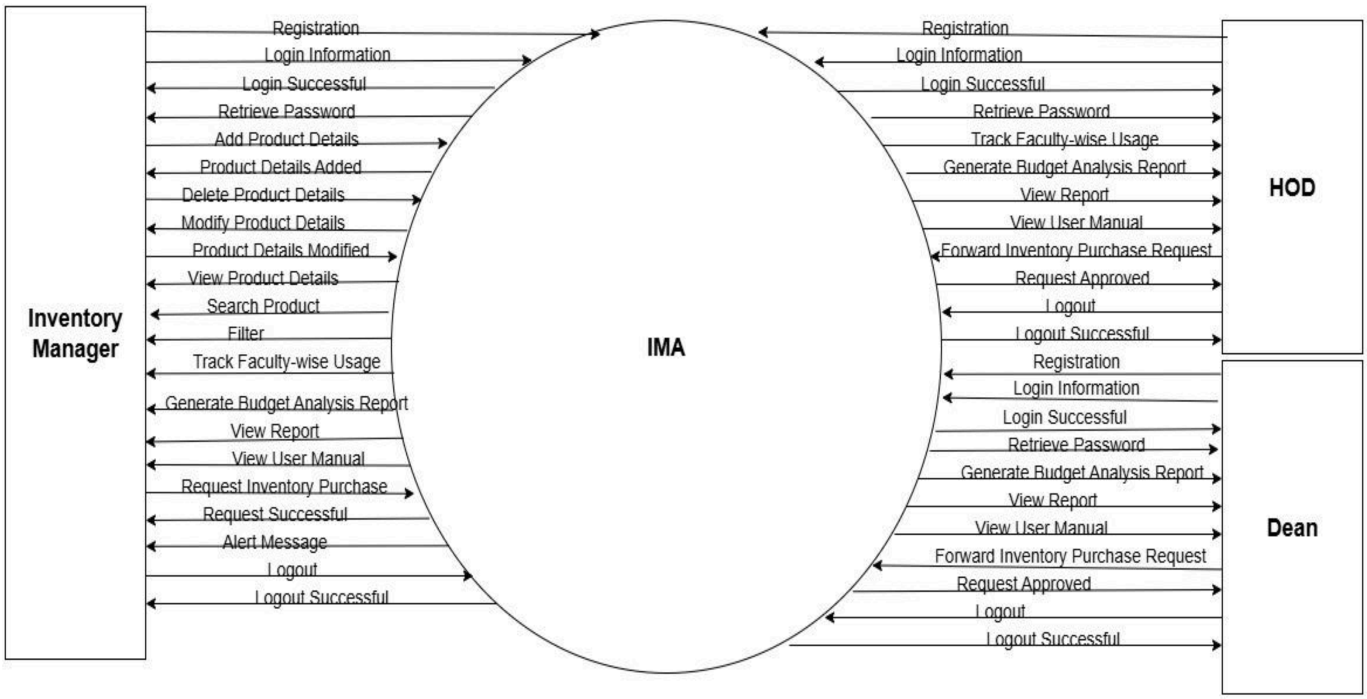
The system has two main user roles:

Inventory Managers and the Administrator(Dean/HOD). Users first go through an Authentication process to log in and access the system's functions. The Dean has access to comprehensive reporting and request approvals, while Inventory Managers can access their specific inventory modules, raise requests and view automatically generated reports.

<u>DFD Level 0:</u>

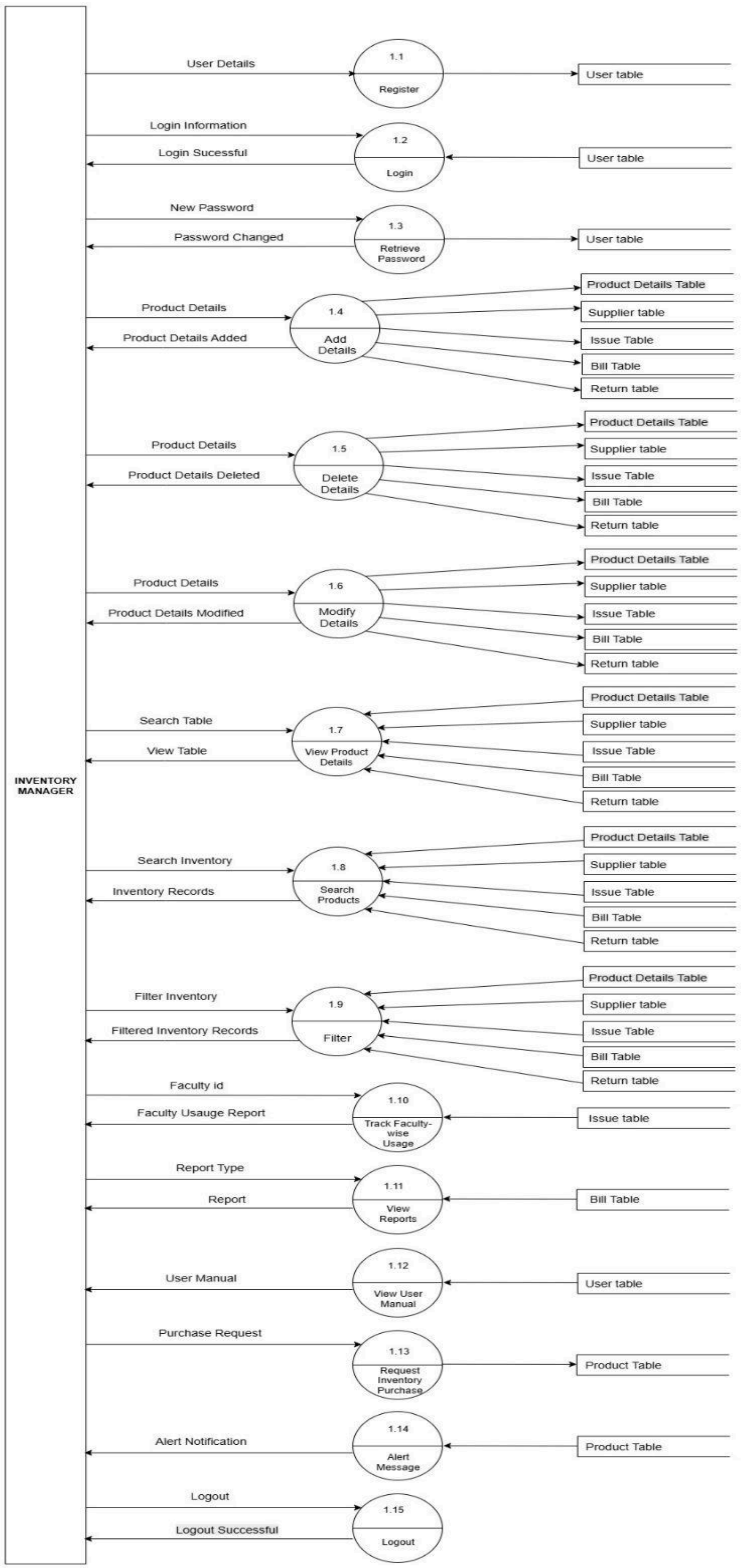
This Data Flow Diagram Level 0, or context diagram, shows the IMA (Inventory Management Application) as a single central process. It illustrates the system's interactions with four external entities: Inventory Manager, HOD (Head of Department), Dean, and Faculty. Each entity sends and receives various data flows to and from the central IMA process. For example, the Inventory Manager can "Add Product Details" and "Generate Budget Analysis Report," while the system returns "Product Details

Added" and the report itself. Similarly, the HOD and Dean both have the ability to "Forward Inventory Purchase Request" to the system and receive a "Request Approved" message in return. This diagram's primary purpose is to provide a high-level overview of the system's boundaries and its key external interactions, without delving into the internal sub-processes.

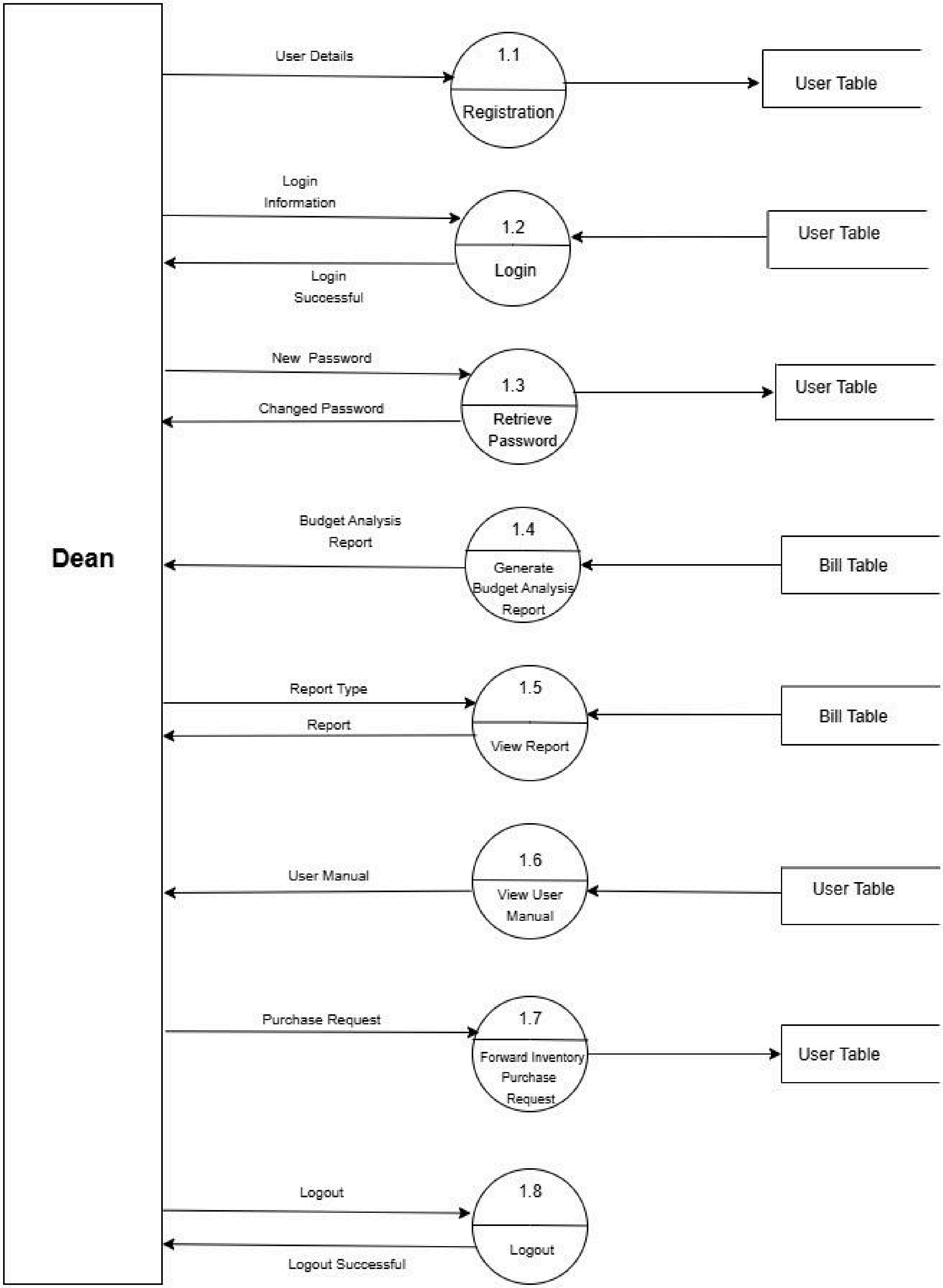


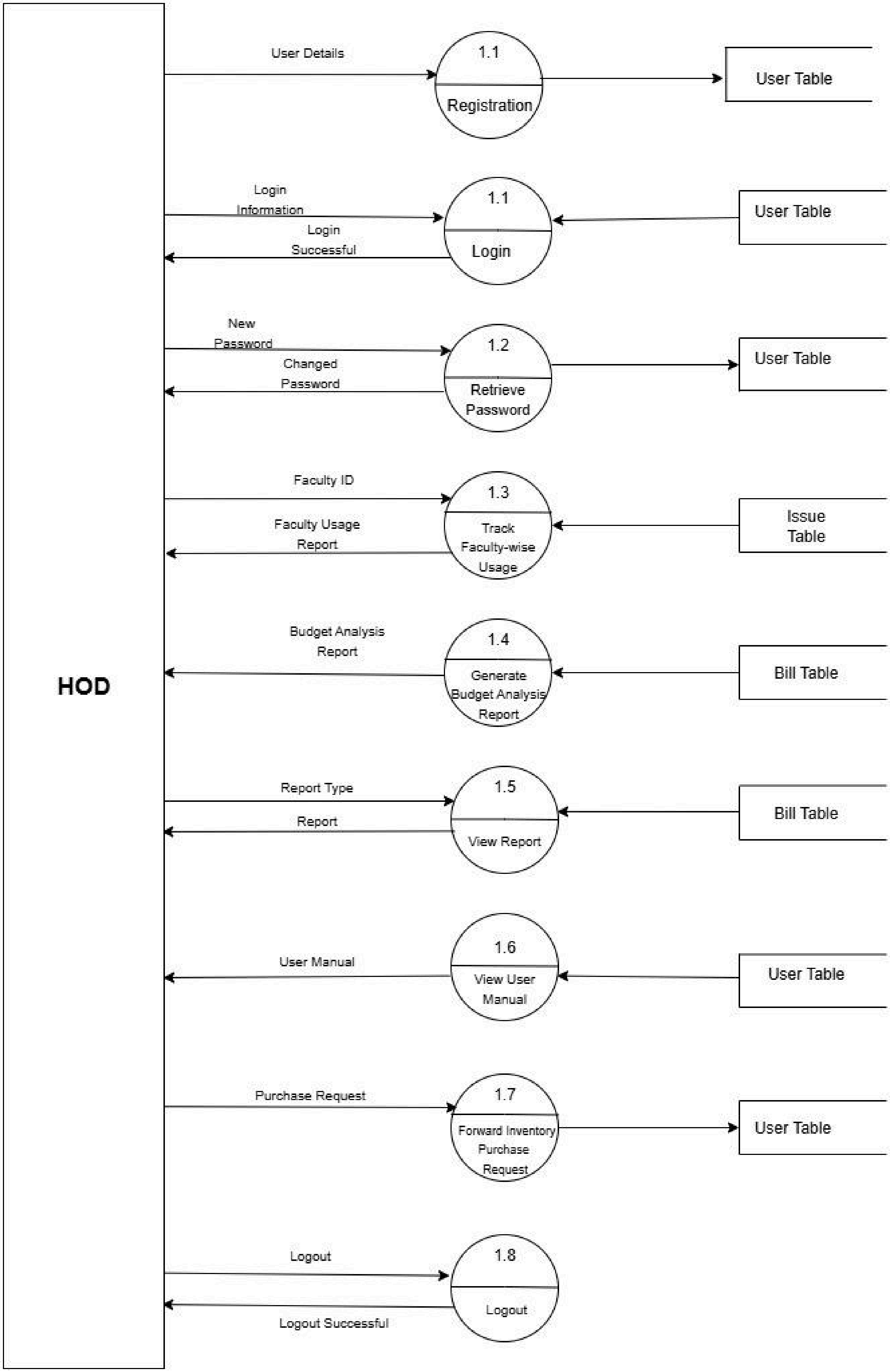
<u>DFD Level 1</u>:

DFD Level 1 diagrams break down the single process from a DFD Level 0, or context diagram, into its major sub-processes. This "explodes" the system to show more detail about how data flows internally. In the provided diagrams, each external entity (Dean, HOD, and Inventory Manager) has its own DFD Level 1, detailing the specific processes and data stores it interacts with. For example, the Dean DFD shows distinct processes like Login (1.2), Retrieve Password (1.3), and Forward Inventory Purchase Request (1.7), each interacting with a data store like the ***User Table*** or ***Bill Table***. Similarly, the Inventory Manager diagram provides a much more granular view of activities like Add Details (1.4), which uses multiple data stores like the ***Supplier, Issue,*** and ***Bill Tables***. This decomposition reveals the main functions of the system and how data moves between processes and data stores to fulfill the system's purpose for each user type.



INVENTORY MANAGER
User Details
1.1
Register
User table
Login Information
Login Sucessful
1.2
Login
User table
New Password
Password Changed
1.3
Retrieve Password
User table
Product Details
Product Details Added
1.4
Add Details
Product Details Table
Supplier table
Issue Table
Bill Table
Return table
Product Details
Product Details Deleted
1.5
Delete Details
Product Details Table
Supplier table
Issue Table
Bill Table
Return table
Product Details
Product Details Modified
1.6
Modify Details
Product Details Table
Supplier table
Issue Table
Bill Table
Return table
Search Table
View Table
1.7
View Product Details
Product Details Table
Supplier table
Issue Table
Bill Table
Return table
Search Inventory
Inventory Records
1.8
Search Products
Product Details Table
Supplier table
Issue Table
Bill Table
Return table
Filter Inventory
Filtered Inventory Records
1.9
Filter
Product Details Table
Supplier table
Issue Table
Bill Table
Return table
Faculty id
Faculty Usauge Report
1.10
Track Faculty-wise Usage
Issue table
Report Type
Report
1.11
View Reports
Bill Table
User Manual
1.12
View User Manual
User table
Purchase Request
1.13
Request Inventory Purchase
Product Table
Alert Notification
1.14
Alert Message
Product Table
Logout
Logout Successful
1.15
Logout


Dean
User Details
1.1
Registration
User Table
Login Information
1.2
Login
Login Successful
User Table
New Password
Changed Password
1.3
Retrieve Password
User Table
Budget Analysis Report
1.4
Generate Budget Analysis Report
Bill Table
Report Type
Report
1.5
View Report
Bill Table
User Manual
1.6
View User Manual
User Table
Purchase Request
1.7
Forward Inventory Purchase Request
User Table
Logout
Logout Successful
1.8
Logout


HOD
User Details
1.1
Registration
User Table
Login Information
Login Successful
1.1
Login
User Table
New Password
Changed Password
1.2
Retrieve Password
User Table
Faculty ID
Faculty Usage Report
1.3
Track Faculty-wise Usage
Issue Table
Budget Analysis Report
1.4
Generate Budget Analysis Report
Bill Table
Report Type
Report
1.5
View Report
Bill Table
User Manual
1.6
View User Manual
User Table
Purchase Request
1.7
Forward Inventory Purchase Request
User Table
Logout
Logout Successful
1.8
Logout

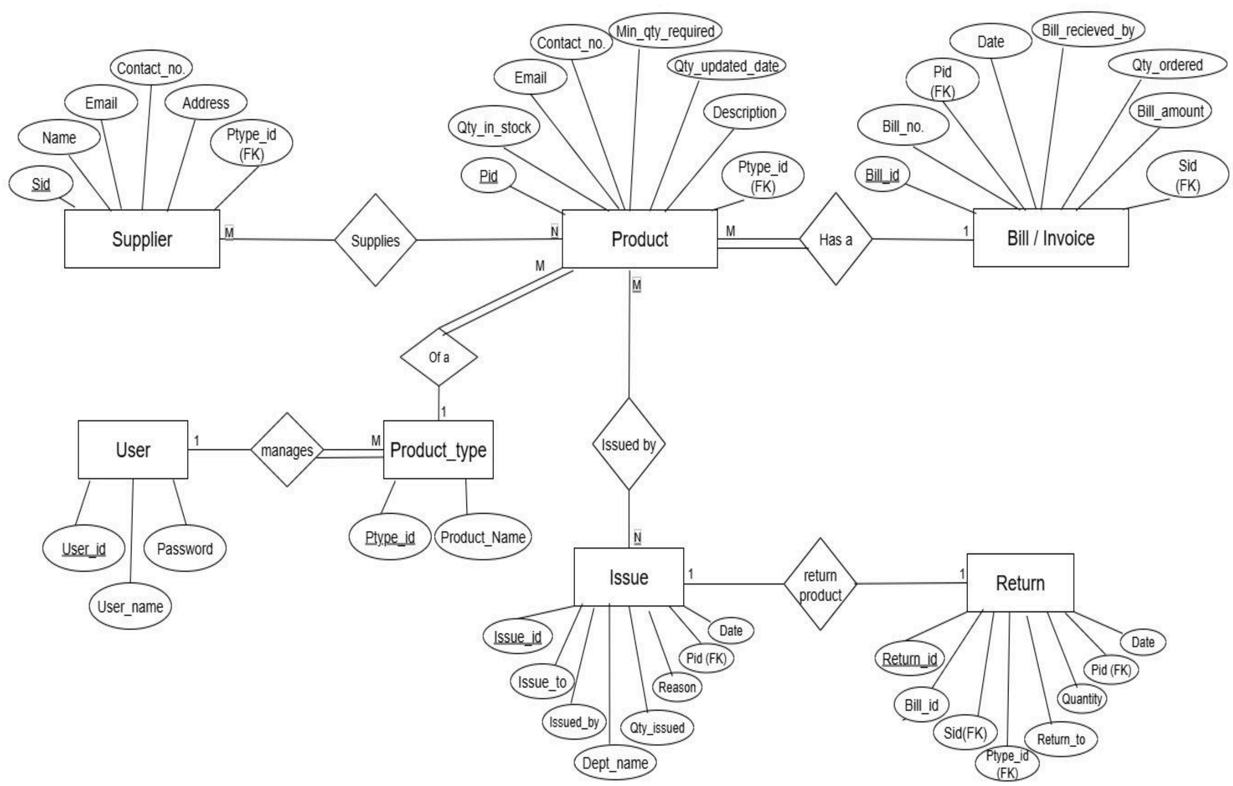
3. Structure and relationships

3.1 Data design

The system will use an SQL database to store all data. The data structures are designed to be normalized to reduce redundancy and ensure data integrity.

3.2 Database description

The database for the Inventory Management System will be implemented using MySQL. The Entity-Relationship (ER) model below illustrates the entities and their relationships.



This ER (Entity-Relationship) diagram models a database for an inventory management system. The core of the system revolves around the Product entity, which is supplied by a Supplier. Each product is linked to a Product_type, which categorizes it.This categorization is managed by a User, who also has access to the system.

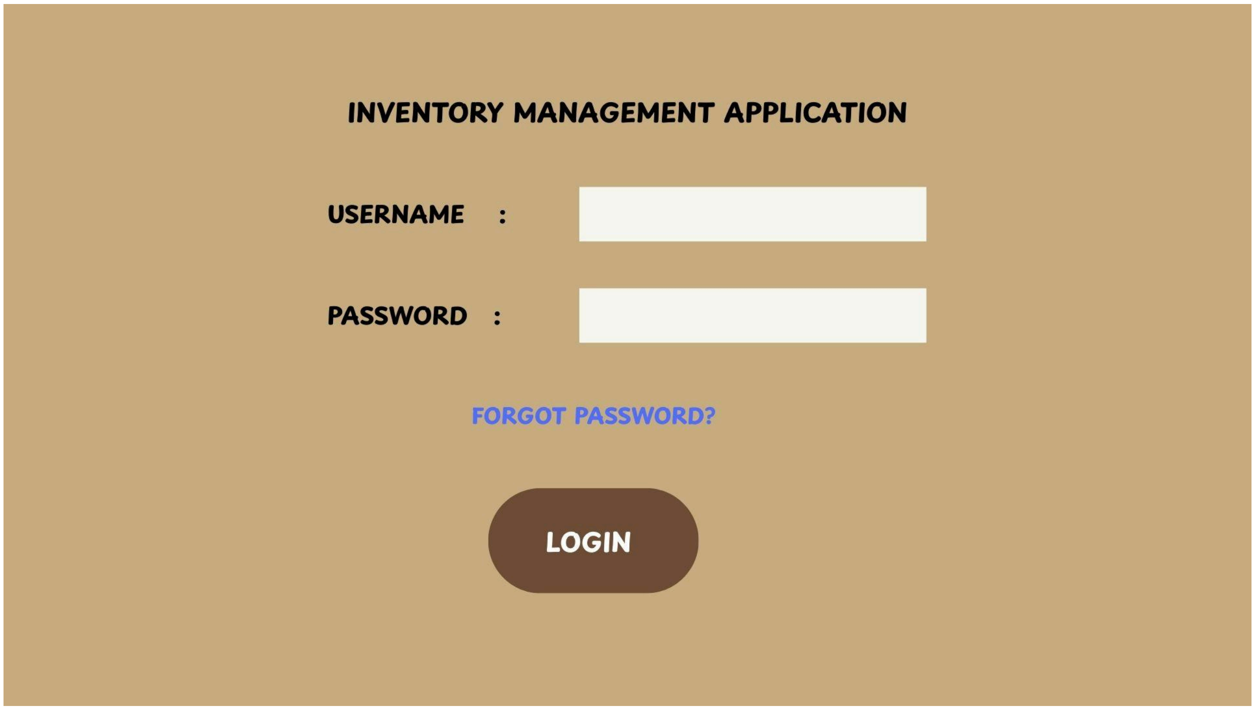
When a product is purchased, it is documented in a Bill, capturing key details like the amount, date, and quantity ordered. The system also tracks internal movements of products using the Issue entity, which records when a product is issued to a specific person or department. Finally, to handle post-sale processes, the Return entity is used to manage returned items, linking each return back to the original bill and supplier. The relationships between these entities (indicated by the lines) are crucial, defining how they interact—for example, a single supplier can provide many products, and a single product can appear on multiple bills. This design ensures that all aspects of the inventory lifecycle, from sourcing to returns, are properly tracked within the application.

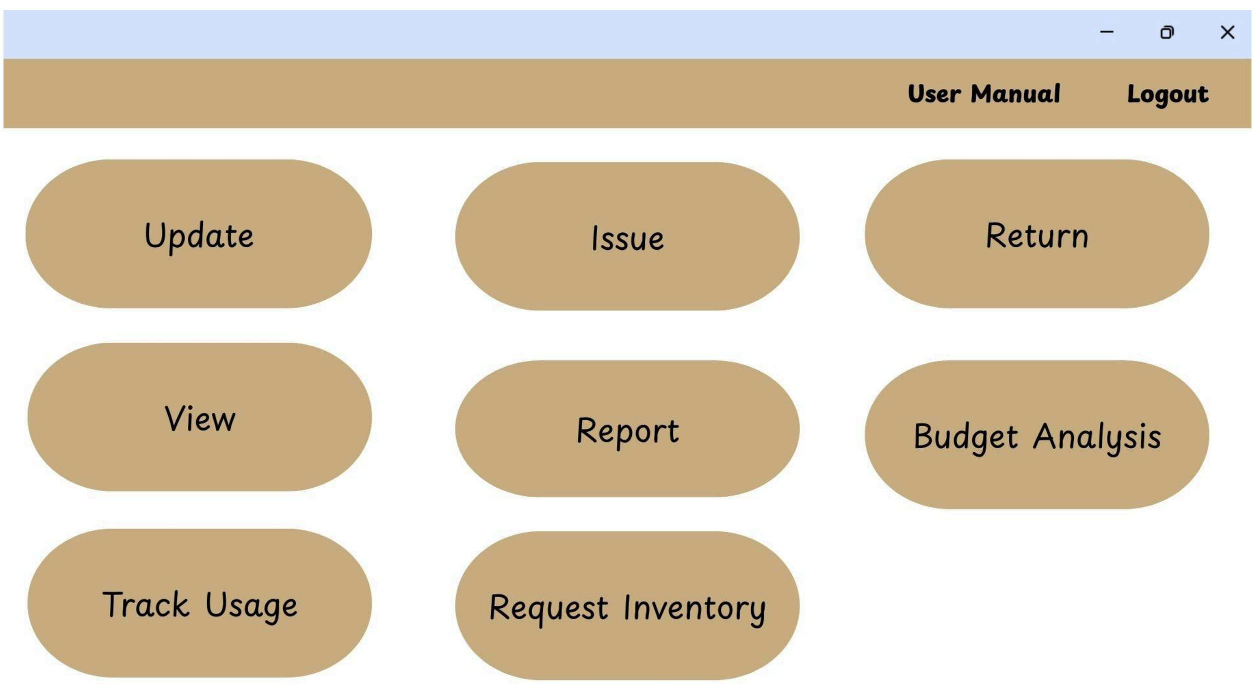
4. User Interface Design

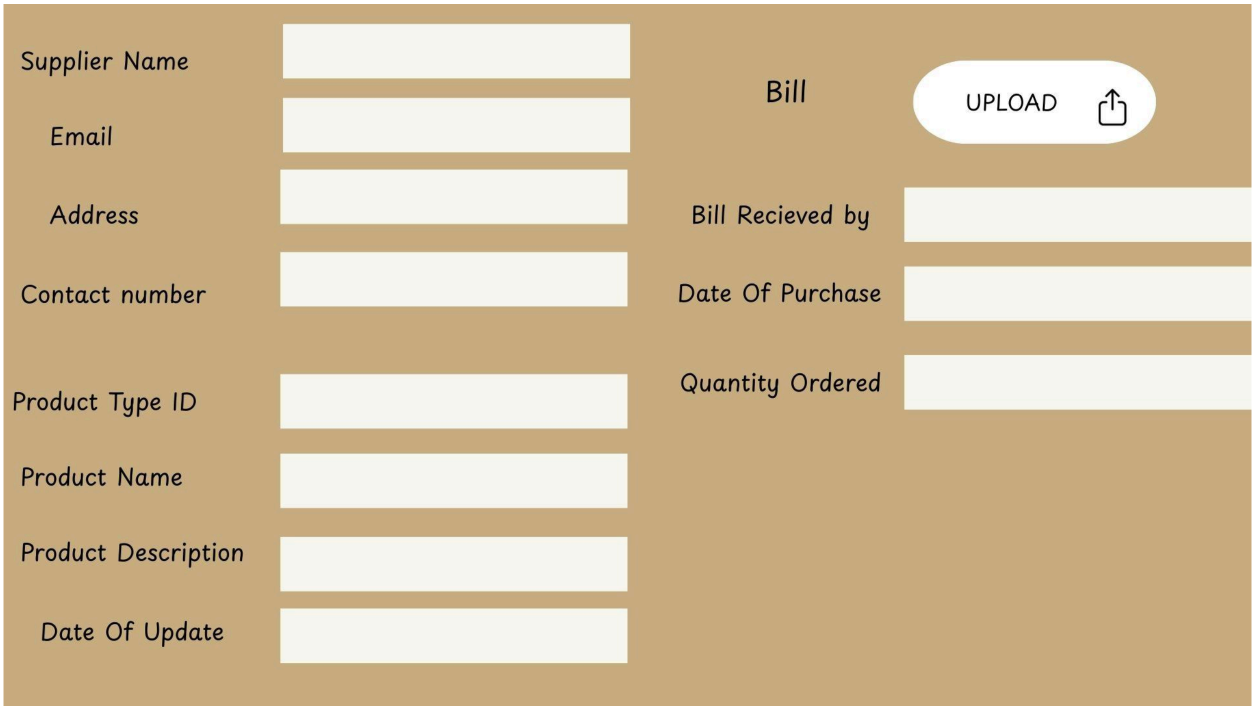
4.1 Description of the user interface

The application will have a graphical user interface (GUI) built with JavaFX, designed for ease of use by office staff with moderate technical skills. The interface will use a consistent, clean layout with navigation menus, forms, and data tables. The design will include standard window controls, validation error messages, and user manual.

4.2 Screen images


INVENTORY MANAGEMENT APPLICATION
USERNAME :
PASSWORD :
FORGOT PASSWORD?
LOGIN


User Manual
Logout
Update
Issue
Return
View
Report
Budget Analysis
Track Usage
Request Inventory



4.3 Objects and actions

The first image represents the Login Interface of the Inventory Management Application, containing objects such as input fields for Username and Password, a Forgot Password link for password recovery, and a Login button to authenticate users and grant access to the system. The second image shows the Main Dashboard after login, which provides different functional actions through button-like objects: Update (to add, delete or modify inventory details), Issue (to allocate items), Return (to process returned items), View (to see existing records), Report (to generate summaries), Budget Analysis (to analyze expenses), Track Usage (to monitor inventory consumption by faculties), and Request Inventory (to request new stock). Additionally, the top bar contains User Manual for guidance and a Logout option to exit the system securely.

5. TYPES OF TESTS

5.1 Unit Tests

Unit tests will be conducted by developers on their individual machines to verify that each component of the application functions as designed. This process will ensure that individual methods, classes, and modules perform their intended tasks correctly, such as data validation for input fields.

5.2 System Tests

The system test environment will be used to test the integration of all modules to ensure they function together seamlessly, as in a typical use case scenario. This will verify end-to-end processes like a user logging in and successfully updating an inventory record and report generation.

5.3 Integration or Regression Tests

A dedicated environment will be used for integration and regression tests. This ensures that new code changes do not adversely affect existing functionalities. This phase is crucial for maintaining the system's stability over multiple development cycles.

5.4 Stress Tests

Stress tests will be performed in an environment identical to the target hosting environment. The goal is to evaluate the system's performance under heavy load, for example, with multiple concurrent users accessing the database. This will help identify potential bottlenecks or deadlocks.

5.5 Acceptance Tests and Staging

Acceptance tests will be conducted to confirm that the system meets the specified requirements and is ready for deployment. This is typically the final testing phase before the application is released to the client.

6. References

Software Requirements Specification for INVENTORY MANAGEMENT SYSTEM, Version 1.0, prepared by CSD026, BANASTHALI VIDYAPITH. This document serves as the foundation for the design outlined in this SDS.

IEEE Recommended Practice for Software Requirements Specifications, IEEE Std 830-1998. This standard was followed in the creation of the SRS and provides a framework for the requirements.

JavaFX Documentation, Oracle, latest release. Used for building the graphical user interface.

Java SE Platform Specification, Oracle, version 17 (LTS). The runtime environment required for the application.

MySQL Official Reference Manual, Oracle, latest edition. The database management system used for the backend.

Departmental guidelines for inventory tracking, faculty of Mathematics and Computing (internal document).